Project: Worked examples
Title: Bolt group (elastic method)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

A pattern of bolts in a plane under in-plane forces (with eccentricity, so torsion), out-of-plane moments and axial load: the shear and tension in every bolt by the elastic (area-weighted) method, the governing bolt, and an optional lever-model prying addition.
Centroid $x_c = \Sigma A x / \Sigma A$. Torsion about the centroid $T = F_y (e_x - x_c) - F_x (e_y - y_c) + T_z$. Polar $J = \Sigma A r^2$. Shear per bolt: $v = A (F/\Sigma A) + T \cdot A \cdot r / J$ (vector sum). Tension per bolt: $t = A N/\Sigma A + A \cdot [(M_x I_y - M_y I_{xy}) y' + (M_y I_x - M_x I_{xy}) x'] / (I_x I_y - I_{xy}^2)$, $I_x = \Sigma A y'^2$, $I_y = \Sigma A x'^2$, $I_{xy} = \Sigma A x'y'$ (this is $M_x A y'/\Sigma A y'^2 + M_y A x'/\Sigma A x'^2$ only for a pattern symmetric about a centroidal axis). Prying: $t_p = t (1 + b/a)$ when a and b are given. Utilisation: the circular interaction $(v/V_R)^2 + (t_p/T_R)^2 \le 1$ (elastic method, Shigley-style) and the EN 1993-1-8 linear form $v/V_R + t_p/(1.4\, T_R) \le 1$, plus each alone; the governing check is the highest.
Ref: EN 1993-1-8:2005 §3.6, §3.8 — Design resistance of individual fasteners; elastic distribution in a bolt group
Ref: AISC Steel Construction Manual Part 7 — Eccentrically loaded bolt groups — elastic method

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
bolts	Bolt positions and areas	4 row(s)		drawing: DRG-BRK-09 rev B	any
bolts[1]		x = 0 mm; y = -100 mm; A = 84.3 mm^2			
bolts[2]		x = 100 mm; y = -100 mm; A = 84.3 mm^2			
bolts[3]		x = 0 mm; y = 100 mm; A = 84.3 mm^2			
bolts[4]		x = 100 mm; y = 100 mm; A = 84.3 mm^2			
F_x	In-plane force, x	0	kN	spec: LOAD-CASE-1	-1000000 … 1000000 kN
F_y	In-plane force, y	0	kN	spec: LOAD-CASE-1	-1000000 … 1000000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
e_x	Point of application, x	0	mm	drawing: DRG-BRK-09 rev B	-100000 … 100000 mm
e_y	Point of application, y	0	mm	drawing: DRG-BRK-09 rev B	-100000 … 100000 mm
T_z	Additional in-plane torque about the centroid	0	kN*m	spec: LOAD-CASE-1	-1000000 … 1000000 kN*m
N	Axial (out-of-plane) load, tension positive	0	kN	spec: LOAD-CASE-1	-1000000 … 1000000 kN
M_x	Out-of-plane moment about x (tension on +y side)	20	kN*m	spec: LOAD-CASE-1	-1000000 … 1000000 kN*m
M_y	Out-of-plane moment about y (tension on +x side)	0	kN*m	spec: LOAD-CASE-1	-1000000 … 1000000 kN*m
V_R	Shear resistance per bolt	40	kN	spec: M12 8.8 shear resistance, EN 1993-1-8 Table 3.4 per project calc	0.1 … 10000 kN
T_R	Tension resistance per bolt	60	kN	spec: M12 8.8 tension resistance per project calc	0.1 … 10000 kN

3. Assumptions

A1. The connected parts are rigid; bolt forces vary linearly with distance from the centroid (elastic method). (Conservative compared with plastic / instantaneous-centre methods.)
A2. Resistances are per bolt as entered, with their source (a design-code capacity or a manufacturer value). (No code capacity is computed here.)
A3. Prying uses a simple lever: the bolt tension rises by the factor (1 + b/a). (An optional, conservative first estimate.)

4. Working

[11] n_b = number of bolts
= 4
= 4
Bolts in the group
[12] x_c = ∑ A x / ∑ A
= 84.3×0 + 84.3×100 + 84.3×0 + 84.3×100 / 337.2
= 50 mm
Centroid x
[13] y_c = ∑ A y / ∑ A
= 84.3×-100 + 84.3×-100 + 84.3×100 + 84.3×100 / 337.2
= 0 mm
Centroid y
[14] T = F_y (e_x − x_c) − F_x (e_y − y_c) + T_z
= 0 × -50 − 0 × 0 + 0
= 0 kN·m
In-plane torque about the centroid
[15] J = ∑ A r²
= about the centroid
= 4215000 mm⁴

Polar second moment of the bolt areas

```
[16] b1 = bolt 1 at (0, -100) mm
= v = (0, 0) kN, t = 0 kN
= 0
```

Interaction utilisation $\sqrt{((v/V_R)^2 + (t/T_R)^2)} = 0$

```
[17] b2 = bolt 2 at (100, -100) mm
= v = (0, 0) kN, t = 0 kN
= 0
```

Interaction utilisation $\sqrt{((v/V_R)^2 + (t/T_R)^2)} = 0$

```
[18] b3 = bolt 3 at (0, 100) mm
= v = (0, 0) kN, t = 50 kN
= 0.83333
```

Interaction utilisation $\sqrt{((v/V_R)^2 + (t/T_R)^2)} = 0.833$

```
[19] b4 = bolt 4 at (100, 100) mm
= v = (0, 0) kN, t = 50 kN
= 0.83333
```

Interaction utilisation $\sqrt{((v/V_R)^2 + (t/T_R)^2)} = 0.833$

```
[20] v_max = governing bolt shear
= bolt 3
= 0 kN
```

Maximum bolt shear

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[21] t_max = governing bolt tension (with prying)
= bolt 3
= 50 kN
```

Maximum bolt tension

```
[22] u_max = √((v/V_R)² + (t/T_R)²)
= √((0/40)² + (50/60)²)
= 0.83333
```

Governing interaction utilisation

```
[23] governing = governing bolt
= bolt 3
= "bolt 3"
```

The bolt with the highest utilisation

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[24] u_v = v_max / V_R
= 0 kN / 40 kN
= 0
```

Shear utilisation of the governing bolt

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[25] u_t = t_max / T_R
= 50 kN / 60 kN
= 0.83333
```

Tension utilisation of the governing bolt

```
[26] u_lin = v/V_R + t/(1.4 T_R)
= 0/40 + 50/(1.4 × 60)
= 0.59524
```

Linear shear–tension interaction (governing bolt)

5. Results

SYMBOL	VALUE	DESCRIPTION
T	0 kN·m	Torque about the centroid
J	4215000 mm⁴	Polar second moment of bolt areas
v_max	0 kN	Maximum bolt shear
t_max	50 kN	Maximum bolt tension

SYMBOL	VALUE	DESCRIPTION
u_max	0.83333	Governing interaction utilisation
governing	"bolt 3"	Governing bolt
u_lin	0.59524	Linear interaction utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Circular shear–tension interaction (governing bolt)	u_max = 0.83333	u_max ≤ 1	0.833	
PASS	Tension alone	u_t = 0.83333	u_t ≤ 1	0.833	
PASS	Linear shear–tension interaction v/V_R + t/(1.4 T_R)	u_lin = 0.59524	u_lin ≤ 1	0.595	EN 1993-1-8 Table 3.4
PASS	Shear alone	u_v = 0	u_v ≤ 1	0	

7. Applicability

No warnings.

8. Verification

Template mech.bolt-group@1.1.0 Content hash fd686de20c6459e9 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0